Sri Datta Mandalapu

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PROFESSIONAL SUMMARY

Software Engineer with 5 years of experience building Java microservices and RESTful APIs using Spring Boot, specializing in asynchronous and event-driven architectures on AWS with Kafka. Skilled in delivering scalable, production-ready services with CI/CD pipelines, Docker/Kubernetes, and serverless deployments. Strong expertise in Relational & NoSQL databases, observability, and security best practices.

TECHNICAL SKILLS

Programming Languages: Java, Python
Web Technologies: React.js, AngularJS, TypeScript, HTML5, CSS3, JavaScript
Frameworks: Spring 3.x, Spring Boot, Hibernate 4.x/5.x
Databases: Oracle 11g, Oracle 12c, MySQL 8+, DynamoDB
Messaging Systems: Kafka, RabbitMQ
Server Management: Apache Tomcat 7.x/9.x, IBM WebSphere 9
Cloud (AWS): EC2, S3, Lambda, API Gateway, DynamoDB, ECS, Fargate, CloudFormation, CloudWatch, X-Ray, IAM, KMS, CloudTrail
Certifications: AWS Certified Cloud Practitioner

EXPERIENCE

Client: Comcast, Philadelphia, PA **Sep 2024 - Present**
Role: Java Developer

- Built and operated the Xfinity Mobile omnichannel API layer, enabling cross-platform onboarding with independently deployable Spring Boot microservices exposing RESTful APIs, improving time-to-market by 25% for certain APIs.
- Delivered customer-facing onboarding in React/TypeScript with strict mode (noImplicitAny, strictNullChecks), Zod-validated React Hook Form wizards, and OpenAPI-generated typed clients, improving completion rate by ~12% and reducing UI defects by ~30%.
- Designed asynchronous, event-driven services using Kafka and RabbitMQ for real-time fraud detection, identity verification, eligibility checks, and payment flows, achieving 100 TPS peak throughput with p99 latency under 300 milliseconds and with 99.9% availability.
- Delivered real-time fraud detection by performing first-party and third-party fraud checks, lowering false positives by 20% and reducing manual review percentage by 50%.
- Implemented credit and exchange eligibility flows with bounded retries and dead-letter handling, improving the approval turnaround by 2%.
- Deployed containerized microservices on ECS with Fargate and API front doors via API Gateway, leveraging Lambda for event triggers and batch tasks, sustaining 99.9%+ availability across Multi-AZ and reducing operational overhead.
- Enhanced observability with centralized logging, metrics, and tracing (e.g., correlation IDs, X-Ray), reducing incident resolution time.
- Optimized data access via Hibernate/JDBC (indexing, batching, pagination) to speed up activation workflows.
- Streamlined releases with Jenkins pipelines and CloudFormation, supporting canary rollouts for safer deployments.

Client: USAA, Plano, Texas **Jan 2021 - Jul 2023**
Role: Java Developer

- Modernized the customer onboarding platform to streamline KYC/CDD checks, reducing cycle time while ensuring compliance.
- Developed microservices for automated customer review scheduling (1/3/5-year) based on risk tiers and policy rules, improving consistency and automation.
- Integrated Kafka to decouple onboarding, risk, and reporting services, enhancing scalability and responsiveness.
- Migrated Angular 5 to 15, delivered reusable components and wizards with Angular Material, improving usability and reducing form errors.

- Implemented failure-to-ticket automation for Control-M payment workflows; attached logs and correlation IDs and auto-assigned tickets to service owners, improving on-call efficiency.
- Built real-time user-group/roles validation for Bill Pay; returned allow/deny decisions instantly and normalized group states by removing inactive entries.
- Automated builds and deployments with Jenkins, Docker, and Gradle; streamlined environment maintenance and release processes.
- Improved code quality with JUnit/Mockito tests, Postman collections, and SonarQube coverage gates.

Client: A3Logics, Hyderabad, India **May 2020 - Dec 2020**
Role: Java Developer

- Developed and maintained insurance domain Java apps using Spring, Maven/Gradle; upgraded JDK versions for performance and stability.
- Deployed Java applications on Apache Tomcat, implemented logging and tracing with Log4j, and used Splunk queries to monitor and analyze application logs.
- Leveraged Java 8 features such as Lambda expressions and Stream API for clean, efficient data processing and functional-style operations.
- Contributed to Agile/Scrum projects (Jira), code reviews, and GitHub workflows; resolved sprint tickets reliably.
- Coordinated in different phases of the product delivery (Analysis, Development, Testing (TDD), Version Control, Deployment, Maintenance, Documentation, Customer Support & Troubleshooting) under Agile software development lifecycle.

EDUCATION

Master of Science, Computer Science **Aug 2023 - May 2025**
University of Houston, Clear Lake, Texas. ***GPA: 3.89***
Related Courses: DBMS, Design and Analysis of Algorithms, Concurrent Programming, Internet Protocols, Machine Learning, AOS, Big Data.

Teaching Assistant **Jan 2024 - May 2025**

- Guided students through lab exercises and supported their learning in Operating Systems and Network Fundamentals courses.
- Assisted in the development and evaluation of projects involving multithreaded applications, such as peer-to-peer communication systems and publish-subscribe messaging projects.
- Evaluated projects, provided feedback, and clarified technical doubts to enhance student comprehension and performance.

PROJECTS

UI for Large Language Models **Jan 2025**

- Built a browser-based interface for locally hosted LLMs (Ollama), replacing command-line interactions.
- Developed a Retrieval-Augmented Generation pipeline with Model Context Protocol server to enhance LLM responses using external tools and resources.
- Created REST APIs with Python FastAPI and Hugging Face Transformers, enabled chat merging and context management for improved model performance.

MCP - Building Rich-Context AI Apps with OpenAI

- Implemented a complete MCP-based pipeline: turned a basic chatbot into an MCP client, developed an MCP server that exposed custom tools, resources, and prompts, and integrated them to allow the chatbot to query external data.
- Implemented an arXiv paper fetcher tool that the model invoked dynamically to enhance responses with research summaries.
- Integrated external data sources and APIs into the server to dynamically enhance LLM responses using real-time contextual information.

ACHIEVEMENTS

- Awarded the Zeon End Grad Fellow Scholarship for academic excellence and outstanding performance.